Sahaj Bhadja

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Summary

Full-stack Software Developer with 2+ years building **React**, **TypeScript**, and **Node.js** apps with Python services, PostgreSQL, payment and business-system **integrations**, AI-powered tools, and CI/CD on AWS and GitHub Actions.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **Python integration pipelines** connecting FastAPI services, PostgreSQL, SQL Server, and cloud storage with incremental loads and reconciliation checks, running 5K+ monthly production jobs at 99%+ reliability.
- Designed normalized **PostgreSQL** schemas with indexing and query tuning for 200+ users and 10K+ records, cutting latency 35%, and isolated failing background jobs to triple throughput across 1K+ AI jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** frontend and Python Flask backend on AWS for upload, rubric, and grading workflows, shipping through **GitHub Actions** CI/CD to serve 150+ instructors and 10K+ daily requests.
- Built **AI-powered tools** that generate rubrics, read handwritten answers with vision OCR, and grade across GPT-4o, Claude, and Qwen-VL with automatic fallbacks, cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Troubleshot and optimized large-scale API **integrations** between business systems with pagination, field filtering, and caching, improving response times 30% for low-bandwidth users and cutting manual data entry 35%.
- Redesigned **AWS** Glue and S3 data workflows, boosting pipeline efficiency 68%, and maintained 20+ automated unit and integration test suites in nightly Jenkins CI across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** React and Node.js business platform for customer records, billing, invoices, and lead management on PostgreSQL, with real-time WebSocket updates syncing 2M+ records across locations.
- Packaged services as **Docker** images through Jenkins CI/CD with Jest and Selenium tests and staged releases, monitored with Sentry, and used analytics to cut resource wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **e-commerce**-style booking, pricing, and coupon APIs in Flask with a React and TypeScript dashboard, validating coupon eligibility and keeping responses under 200ms across 50K+ booking records.
- Integrated card, wallet, and PayPal **payment** providers through idempotent webhooks for refunds and disputes, cutting manual refund handling about 70%, and presented the architecture to the client.

Skills

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS, Java
Frontend & AI: React, Next.js, TailwindCSS, React Native, WebSockets, Responsive Design, LLM APIs, OCR
Backend: Node.js, Express.js, Flask, FastAPI, Django, REST APIs, Microservices, Webhooks
Data & Cloud: PostgreSQL, MySQL, SQL Server, AWS (S3, RDS, EC2, Glue), Docker, Firebase
Quality & Tools: Jest, Selenium, PyTest, CI/CD, GitHub Actions, Jenkins, Git, Code Reviews, Sentry, Stripe, Agile

Projects

Custom Product Configurator App | *Remix, React, TypeScript, Prisma, PostgreSQL, three.js*

- Designed a **Shopify**-embedded Remix app with a **three.js** configurator where shoppers adjust size and finish on a 3D model, see live pricing, and save choices to orders through the Admin GraphQL API.

Automated Product Render Pipeline | *Python, Blender, Docker, AWS ECS, S3, CloudWatch*

- Designed a **Python** pipeline that renders every product variant in headless **Blender** inside Docker on AWS ECS, uploads images to S3, syncs them to store listings, and alerts on failures through CloudWatch.

PiXMod Live | *Next.js, Stripe, SAM, Stable Diffusion, CLIP, MongoDB*

- Built an AI-powered **image** SaaS with SAM segmentation, Stable Diffusion editing and restoration, CLIP search, and Stripe subscription billing behind a responsive Next.js frontend, raising premium conversion 18%.